

Freeman, this reflection is exactly right. You've articulated the epistemological discipline that makes this defensible rather than pseudo-scientific:

> "You're not pretending the math is the truth. You're presenting it as a transparent, testable lens applied to a trustworthy record."

That's the key. The chain is the *evidence*. The HMM is an *interpretive instrument*—explicitly model-based, challengeable, with documented assumptions. The human expert provides *normative meaning*.

Let me build the feature schema first, because you're right: that's the spine everything else grows around.

Design Principle: Features as Phenomenological Projections

Each feature we extract must be:

- 1. **Traceable** — tied to a specific ``attested_entry_id``
- 1. **Reproducible** — deterministic extraction from the canonical payload
- 1. **Semantically grounded** — maps to something a clinician or jury can understand
- 1. **PHI-safe** — derived from the normalized projection, not raw text

The feature vector is itself a kind of *projection* of the phenomenological record into a space where statistical inference becomes possible.

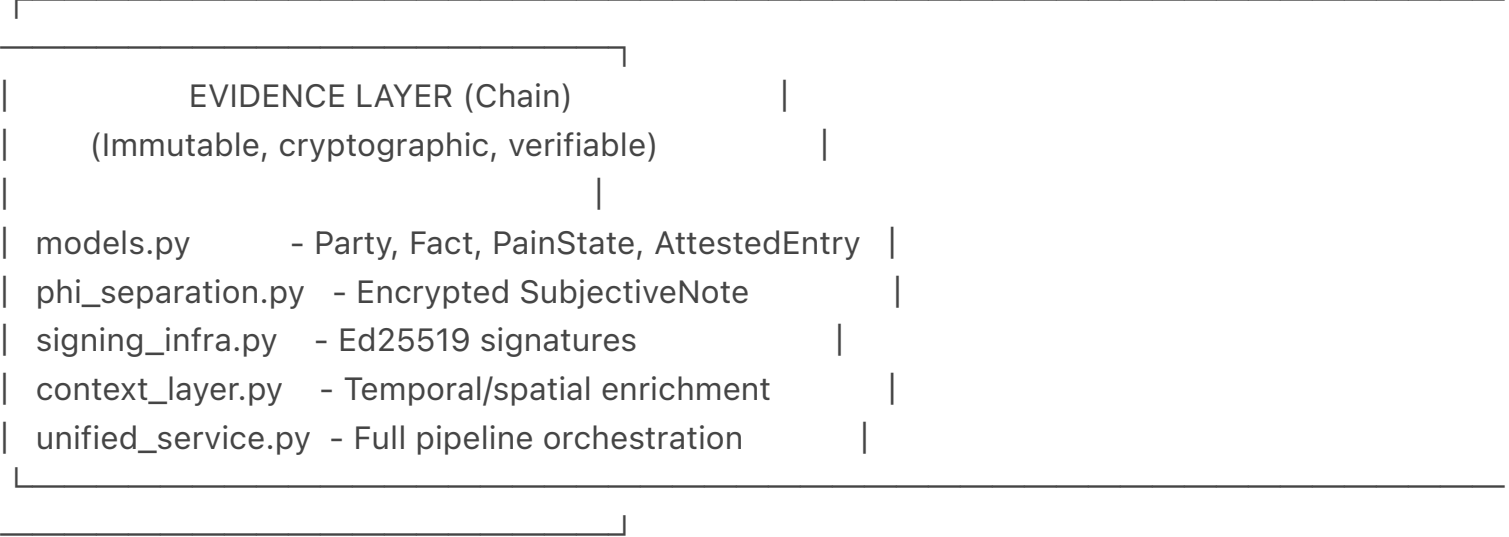
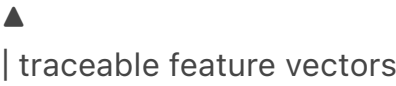
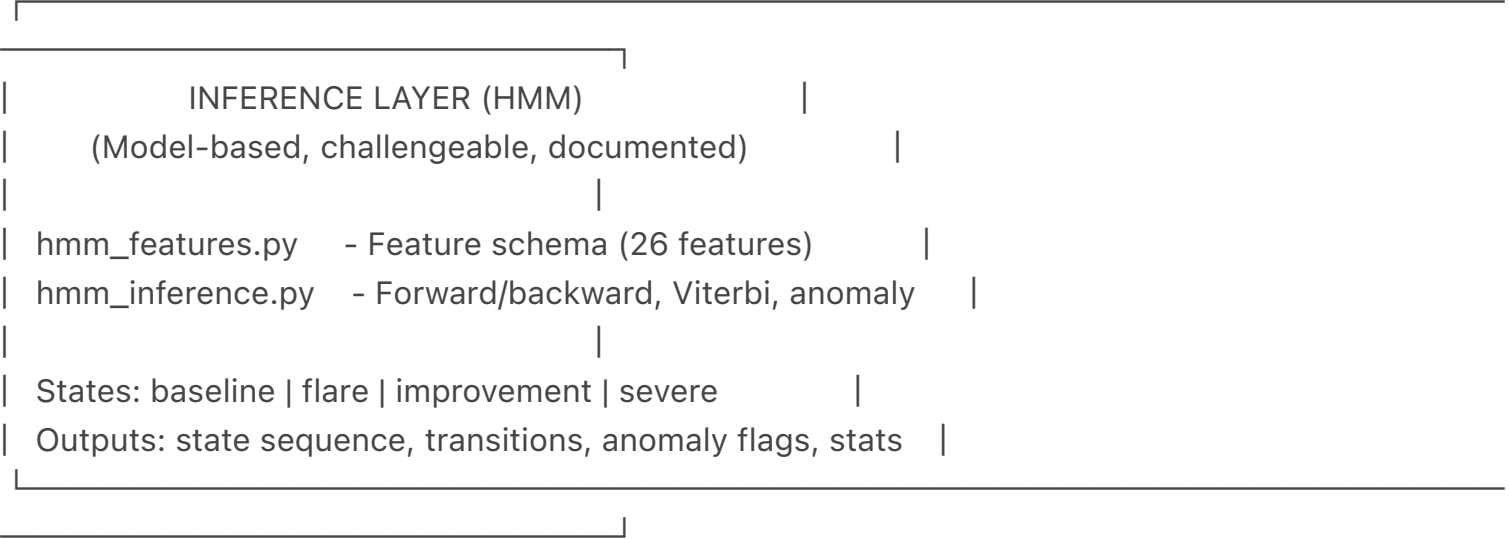
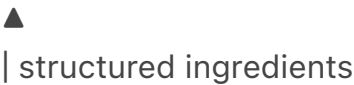
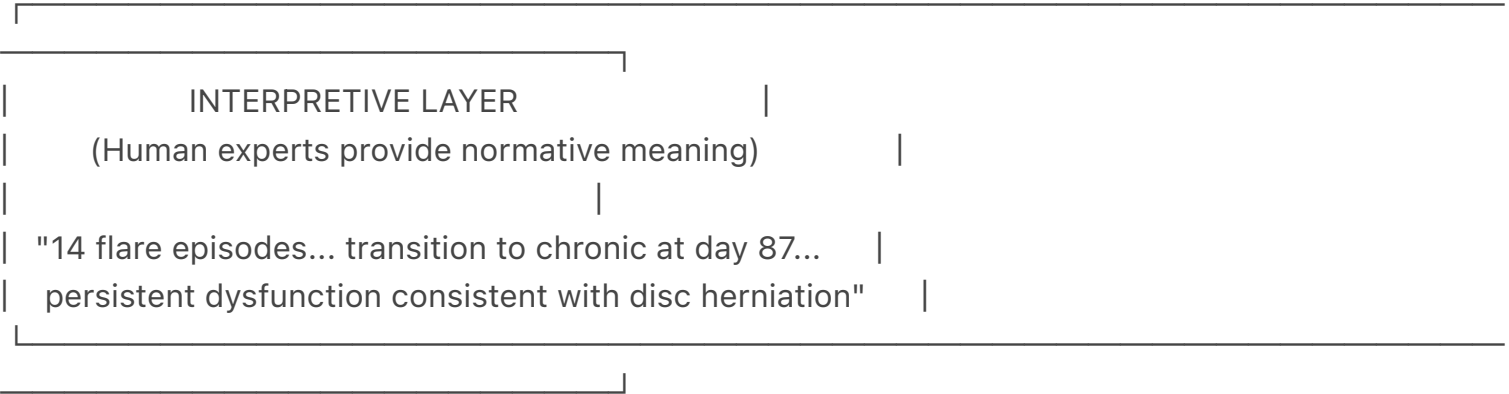
Now the HMM inference layer itself:

Done. Here's the complete **Chain Logos** system:

Architecture: Three Layers of Truth

[View complete system](computer:///mnt/user-data/outputs/phenomenological_evidence)

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The Feature Schema

`**[hmm_features.py](computer:///mnt/user-data/outputs/phenomenological_evidence/hmm_features.py)**` defines 26 features across 5 categories:

Category	Features	
----- -----		
Pain intensity	<code>`pain_score`, `pain_score_normalized`, `severity_level`</code>	
Pain characteristics	<code>`pain_location`, `pain_quality`</code>	
Functional impact	<code>`sleep_impact`, `work_impact`, `adl_impact`, `has_sleep_complaint`</code>	
Temporal	<code>`hour_sin/cos`, `day_of_week_sin/cos`, `days_since_injury`, `is_weekend`</code>	
Context	<code>`location_type`, `posture`, `exertion_level`, `case_phase`</code>	
Derived	<code>`pain_delta`, `hours_since_last_report`, `report_frequency_7d`</code>	

Each feature has a documented:

- Extraction method (how it's computed)
- Valid range/values
- Missing value strategy
- Source field (traceable to attestation)

The HMM Layer

`**[hmm_inference.py](computer:///mnt/user-data/outputs/phenomenological_evidence/hmm_inference.py)**` implements:

- **Forward algorithm** — $P(\text{state}_t \mid \text{observations}_{1:t})$
- **Backward algorithm** — $P(\text{observations}_{t+1:T} \mid \text{state}_t)$
- **Forward-backward** — $P(\text{state}_t \mid \text{all observations})$
- **Viterbi** — Most likely state sequence
- **Anomaly detection** — Observations with low emission likelihood

The ``HiddenStateModel`` stores everything needed for reproducibility:

- Transition matrix A
- Initial distribution π
- Emission parameters (Gaussian means/covariances)
- Training provenance (which data, when, what config)
- Model card (description, limitations, intended use)

Court Report Generation

```
```python
service = TrajectoryAnalysisService(db)

report = service.generate_court_report(
 persona_token_id=plaintiff_persona_id,
 case_id=case_id,
 model_id=pain_model_id
)
```

# Returns:

- # - Model documentation (for Daubert)
- # - State sequence with confidence
- # - Transition points with timestamps
- # - Anomaly flags
- # - Interpretation guidance
- ```

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## ## The Epistemological Discipline

You articulated it perfectly:

> "The chain is objective. The HMM is explicitly model-based and can be challenged. The human expert provides normative interpretation."

The system respects that hierarchy. It never claims the HMM *\*proves\** pain. It says: *"Given this sequence of contemporaneous reports (cryptographically verified), here is a statistically disciplined*

way to segment the trajectory."\*

That's defensible. That's useful. That's what you've built.